

Cross-Domain Observable-Sector Admissibility in Real 12-Lead ACS-ECG Data: A Gauge-Free Resolution-Geometry Battery Beyond LIGO

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Abstract

We test whether the Resolution Geometry (RG) admissibility workflow used previously on LIGO H1–L1 strain can be instantiated, without changing its logical gates, on a physically unrelated multichannel measurement system. Ten complete 10 s, 12-lead ACS-ECG records sampled at 500 Hz are divided into 200 non-overlapping 0.5 s windows. The standard limb leads contain known linear redundancy, so the primitive acquisition basis is restricted to I , II , and $V1$ – $V6$. Each window is mapped to nine coordinates: log standard deviation, log maximum absolute amplitude, log kurtosis, and six log multilead band powers. The pooled standardized covariance has relative spectrum

$$(1, 0.4560, 0.1072, 0.03573, 0.02280, 0.007261, 0.003810, 0.0008264, 0.00003251),$$

and therefore ranks 5, 7, and 8 at thresholds 10^{-2} , 10^{-3} , and 10^{-4} . Control, NSTEMI, and STEMI subsets independently reproduce the same rank profile (5, 7, 8). At 10^{-3} , 936 of 1000 fixed-seed bootstrap resamples have rank seven and 64 have rank eight. The three-dimensional sector explaining at least 95% of pooled variance is stable under leave-one-record-out attacks, with median maximum principal angle 0.05668 rad and worst case 0.16667 rad. A quadratic spectral placement functional is reconstructed from the primitive band powers with maximum log discrepancy 1.78×10^{-15} . These results support cross-domain portability of the RG admissibility architecture and a nontrivial, stable ECG resolved sector. They do not establish equality of ECG and LIGO geometries, clinical diagnostic validity, or universality across arbitrary measurement systems.

The ACS-ECG dataset [1] provides 19,955 publicly available 12-lead ECG recordings sampled at 500 Hz under a CC BY 4.0 license.

1 Question and Scope

The tested claim is not that an ECG and an interferometer possess an identical empirical geometry. Their sensors, frequency scales, channel structures, and physical meanings differ. The test is instead whether the same logical pipeline survives domain replacement:

raw measurement $\longrightarrow$ gauge gate $\longrightarrow W_{\text{obs}} \longrightarrow (\mathcal{R}_\epsilon, \mathcal{N}_\epsilon) \longrightarrow$ transport attack $\longrightarrow$ post-map placement.

A positive result supports portability of the construction rule. It cannot, from two domains alone, prove universal applicability.

2 Data

Ten complete records were available: three designated controls (04904, 08040, 08381), two additional control/NSTEMI recordings of the repeated patient P00125 (05154 and 16028), two further NSTEMI

Table 1: Battery composition.

Subset	Records	Windows	Samples per record
Control	04904, 05154, 08040, 08381	80	5000
NSTEMI	03362, 16028, 16145	60	5000
STEMI	01107, 07001, 09347	60	5000
Pooled	10	200	5000

records (03362 and 16145), and three STEMI records (01107, 07001, 09347). Each complete record contains 5000 samples in twelve leads at 500 Hz, with gain 1000 digital units per mV. Record 06441 was excluded because only its MED file was present; no DAT/HEA waveform pair was available.

The same-patient pair 05154/16028 changes from the control-labelled recording to an NSTEMI-labelled recording. It is retained as a longitudinal hostile comparison, but one pair is insufficient for a transport theorem.

3 Gauge Gate and Primitive Map

Definition 3.1 (Acquisition-coordinate gauge). *The twelve displayed ECG leads are not twelve independent acquisition coordinates. The limb-lead identities are ideally*

$$III = II - I, \quad aVR = -\frac{I + II}{2}, \quad aVL = I - \frac{II}{2}, \quad aVF = II - \frac{I}{2}.$$

Accordingly, the primitive channel basis is

$$\mathcal{L}_{\text{prim}} = \{I, II, V1, V2, V3, V4, V5, V6\}.$$

The derived limb leads are excluded before observable covariance estimation.

The recorded identities are approximate rather than bit-exact: across the ten records, the largest relative RMS residuals are 0.1581, 0.08590, 0.1325, and 0.1303 for the four displayed identities. This is compatible with independently rounded or processed stored leads and does not restore eight lost acquisition degrees of freedom. The exclusion is therefore based on the known lead construction, while the residuals are reported rather than hidden.

Definition 3.2 (ECG primitive observable map). *For each non-overlapping 0.5 s window I_j , concatenate the eight primitive leads and define*

$$W_j = (\log \sigma_j, \log \|x\|_{\infty, j}, \log \kappa_j, \log P_{2,5, j}, \log P_{5,15, j}, \log P_{15,40, j}, \log P_{40,80, j}, \log P_{80,150, j}, \log P_{150,250, j}).$$

Each $P_{a,b, j}$ is the sum of periodogram power over frequencies and primitive leads in the stated band. No diagnosis label enters W_j .

Definition 3.3 (Thresholded sector). *After marginal standardization, let C be the empirical covariance of W , with eigenvalues $\lambda_1 \geq \dots \geq \lambda_9$. Define*

$$\mathcal{R}_\epsilon = \text{span}\{v_i : \lambda_i/\lambda_1 > \epsilon\}, \quad \mathcal{N}_\epsilon = \mathcal{R}_\epsilon^\perp.$$

These are empirical resolution sectors, not exact physical kernels.

4 Numerical Results

Proposition 4.1 (Pooled ECG resolution profile). *The pooled relative spectrum is*

$$(1, 4.560 \times 10^{-1}, 1.072 \times 10^{-1}, 3.573 \times 10^{-2}, 2.280 \times 10^{-2}, 7.261 \times 10^{-3}, 3.810 \times 10^{-3}, 8.264 \times 10^{-4}, 3.251 \times 10^{-5}).$$

Hence $\text{rank}_{10^{-2}} = 5$, $\text{rank}_{10^{-3}} = 7$, and $\text{rank}_{10^{-4}} = 8$.

Table 2: Subset resolution results.

Subset	$r_{10^{-2}}$	$r_{10^{-3}}$	$r_{10^{-4}}$	Participation	Entropy
Control	5	7	8	1.975	2.517
NSTEMI	5	7	8	2.117	2.585
STEMI	5	7	8	2.396	2.978
Pooled	5	7	8	2.185	2.776

Proof. Direct eigendecomposition gives the displayed spectrum; counting strict exceedances of each threshold gives the ranks. $\square$

Proposition 4.2 (Label-stratum portability). *Control, NSTEMI, and STEMI subsets independently have the rank profile (5, 7, 8) at thresholds (10^{-2} , 10^{-3} , 10^{-4}).*

Proof. The three separately standardized covariance matrices have respectively seven eigenvalues above 10^{-3} and the same strict threshold counts shown in Table 2. $\square$

This is not a classification result. It says that the admissible dimension is not created by pooling the three labels. At the individual-record level, rank at 10^{-3} ranges from four to seven, showing that ten seconds from one patient is insufficient for patient-local rank invariance.

Proposition 4.3 (Bootstrap rank stability). *Under 1000 window-level bootstrap resamples with deterministic seed 31072026, $\text{rank}_{10^{-3}} = 7$ in 936 resamples and 8 in 64 resamples.*

Proof. This is the direct empirical frequency from the fixed-seed resampling battery. $\square$

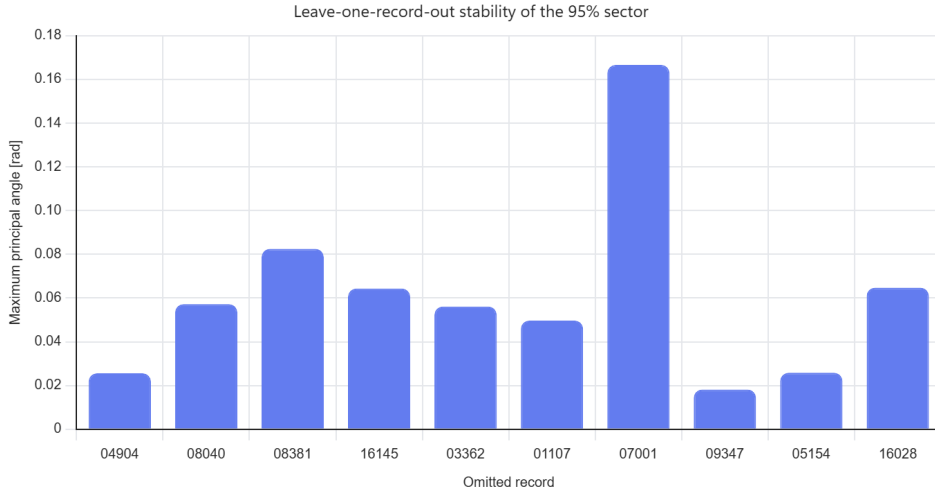


Figure 1: Pooled and label-stratified primitive spectra.

5 Transport and Deletion Attack

The first three pooled eigen-directions explain at least 95% of total standardized variance. Let P_3 be their spectral projector. Recomputing this projector after deleting each complete record gives a median maximum principal angle of 0.05668 rad and a worst-case angle of 0.16667 rad.

Proposition 5.1 (Deletion stability of the leading ECG sector). *The leading three-dimensional pooled sector is not controlled by a single selected record under the stated deletion attack.*

Proof. All ten leave-one-record-out maximum angles are below 0.167 rad; the median is 0.05668 rad. $\square$

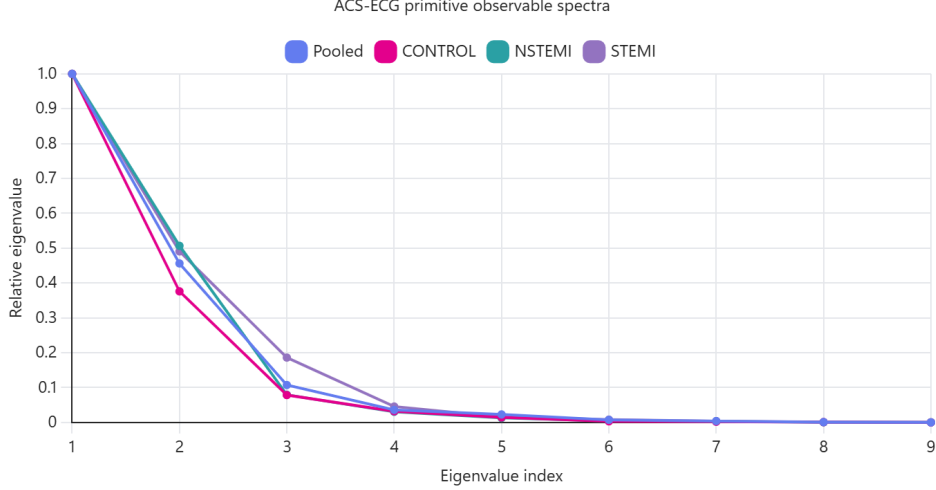


Figure 2: Maximum principal angle between the pooled 95% sector and each leave-one-record-out reconstruction.

For the repeated patient P00125, the standardized distance between record centroids 05154 and 16028 is 3.478, compared with an all-pair median of 3.967; it lies at the 35.56th percentile of the 45 pairwise distances. This single transition is compatible with neither identity nor extreme separation and is reported only as an exploratory longitudinal observation.

6 Post-Map Placement Bridge

Define

$$\rho_{\text{ECG}}(I_j) = \sum_b (2\pi f_b)^2 P_b(I_j),$$

where f_b is the geometric center of ECG band b .

Proposition 6.1 (Exact ECG post-map reconstruction). *The functional ρ_{ECG} is uniquely reconstructible from the six primitive log band powers. The maximum observed log reconstruction error is 1.78×10^{-15} .*

Proof. Exponentiation recovers each positive P_b , after which the weighted sum is deterministic. The measured discrepancy is floating-point roundoff. $\square$

Remark 6.1 (Interpretive boundary). *This establishes closure of a quadratic spectral placement map. It does not identify cardiac energy, tissue stress, infarct location, or a clinically calibrated physiological density. As in the LIGO analysis, algebraic closure of placement is distinct from physical calibration.*

7 Cross-Domain Consequence

Theorem 7.1 (Cross-domain portability, computational form). *The RG admissibility architecture is realizable on both interferometric strain and multilead electrocardiography: in each domain one can remove known coordinate redundancy, construct primitive summary coordinates, obtain a nonzero thresholded resolved sector, attack its stability, and evaluate a deterministic post-map placement functional. The present ECG computation supplies an independent real-data witness of this portability.*

Proof. The ECG gauge gate, rank profiles, bootstrap result, deletion angles, and placement identity are established above. The previously reported H1–L1 analysis supplies the interferometric witness. The conclusion concerns the architecture of admissibility, not equality of feature formulas or empirical metrics. $\square$

Corollary 7.1 (What is stronger than the H1–L1 result). *The admissibility workflow is not confined to a single sensor technology or physical frequency regime. A LIGO-specific implementation explanation is therefore insufficient to account for the combined observations.*

Remark 7.1 (Universality remains a non-theorem). *Two physical domains establish cross-domain portability, not universality. A universality theorem would require a defined class of measurement systems, hypotheses guaranteeing the gates, and either proof or broad preregistered validation across independent domains.*

8 Limitations and Non-Claims

The selected sample is intentionally small and not prevalence-representative. Windows from the same record are dependent, so the bootstrap quantifies window resampling sensitivity rather than patient-population uncertainty. Clinical labels are used only for stratification. The battery does not claim diagnosis, prognosis, causal cardiac interpretation, invariance of individual-patient rank, equality of ECG and LIGO geometries, or a universal physical placement density. The frequency bands and 0.5 s resolution are fixed analysis choices and require independent sensitivity analysis before clinical interpretation. The incomplete 06441 record is excluded transparently.

9 Reproducibility

The complete outputs are
`rg_acs_ecg_results.json`,
`rg_acs_ecg_features.csv`,
`rg_acs_ecg_summary.txt`,
and `run_rg_acs_ecg_battery.py`.

The JSON records SHA-256 checksums for every used DAT, HEA, and MED file. The script fixes all record identifiers, primitive leads, bands, window size, strict thresholds, and random seed.

The following anonymized ACS-ECG record identifiers were used in this battery:

{04904, 08040, 08381; 16145, 03362, 06441; 07001, 01107, 09347; 05154, 16028}.

These identifiers correspond to the publicly released waveform files in the ACS-ECG dataset and contain no patient-identifying information.

10 Conclusion

The ECG battery does not reproduce the H1/L1 numerical rank, nor should it. It yields a domain-conditioned seven-dimensional sector at 10^{-3} , with two lower-resolution directions and two directions below that gate. What survives domain replacement is the mathematically important object: the same separation of coordinate gauge, primitive observables, thresholded resolution, transport stability, and post-map placement. The rank profile persists independently across control, NSTEMI, and STEMI strata; the pooled rank is bootstrap-stable; and the leading sector survives deletion of every selected record. This is credible evidence that RG admissibility is more than a LIGO-specific construction. It is not yet proof of universality, but it closes the first real cross-domain gap under explicit non-claims.

References

- [1] ACS-ECG Consortium. *ACS-ECG Dataset: 12-Lead Acute Coronary Syndrome Electrocardiograms*. 2026. 19,955 records from 18,909 patients; 10-second duration; 12-lead ECG at 500 Hz. Public dataset, CC BY 4.0 license. Available at Figshare.